MODULE III: ARRAYS & STRINGS

BASIC CONCEPT OF ARRAYS

• Consider a situation, where we need to store 5 integer numbers.
• If we use simple variable and data type concepts, then we need 5 variables of int data type and program will be something as follows:

```
#include<stdio.h>
void main()
{
  int  number1;
  int  number2;
  int  number3;
  int  number4;
  int  number5;

  number1 = 10;
  number2 = 20;
  number3 = 30;
  number4 = 40;
  number5 = 50;

  printf( "number1: %d \n", number1);
  printf( "number2: %d \n", number2);
  printf( "number3: %d \n", number3);
  printf( "number4: %d \n", number4);
  printf( "number5: %d ", number5);
}
```

Output:

```
number1: 10
number2: 20
number3: 30
number4: 40
number5: 50
```

• It was simple, because we had to store just 5 integer numbers. Now let's assume we have to store 5000 integer numbers, so what is next? Are we going to use 5000 variables?
• To handle such situation, C language provides a concept called the array.
• Some examples where arrays can be used are
    1) List of temperatures recorded every hour in a day, or a month, or a year
    2) List of employees in an organization
    3) List of products and their cost sold by a store
    4) Test scores of a class of students

ARRAY

• Array is a collection of elements of same data type.
• The elements are stored sequentially one after the other in memory.
• Any element can be accessed by using
    → name of the array
    → position of element in the array
• Arrays are of 2 types:
    1) Single dimensional array
    2) Multi dimensional array

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SINGLE DIMENSIONAL ARRAY

- A single dimensional array is a linear list consisting of related elements of same type.
- In memory, all the elements are stored in continuous memory-location one after the other.

Declaration of Single Dimensional Arrays

- The syntax is shown below:

  data_type array_name[array_size]

  where data_type can be int, float or char

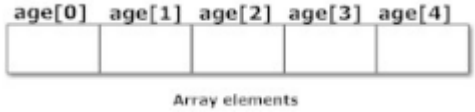
  array_name is name of the array

  array_size indicates number of elements in the array

- For ex:

  int age[5];

- The above code can be pictorially represented as shown below:



Array elements

- Note that, the first element is numbered 0 and so on.
- Here, the size of array 'age' is 5 times the size of int because there are 5 elements.

Storing Values in Arrays

- The values can be stored in array using following three methods:
  1) Initialization
  2) Assigning values
  3) Input values from keyboard

Initialization of One-Dimensional Array

- The syntax is shown below:

  data_type array_name[array_size]={v_1,v_2,v_3};

  where v_1, v_2, v_3 are values

- For ex:

  int age[5]={2,4,34,3,4};

- The above code can be pictorially represented as shown below:



Initialization of one-dimensional array

- Example: Program to illustrate the initialization of one-dimensional array.

```
#include<stdio.h>
void main()
{
        int age[5]={2,4,34,3,4};
        printf("Value in array age[0] : %d \n", age[0]);
        printf("Value in array age[1] : %d \n", age[1]);
        printf("Value in array age[2] : %d \n", age[2]);
        printf("Value in array age[3] : %d \n", age[3]);
        printf("Value in array age[4] : %d ",age[4]);
}
```

Output:

```
Value in array age[0] :2
Value in array age[1] :4
Value in array age[2] :34
Value in array age[3] :3
Value in array age[4] :4
```


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Assigning values to One-Dimensional Array

- For ex:

```
int age[5];
age[0]=2;     //value 2 stored in array 'age' at position 0
age[1]=4;     //value 4 stored in array 'age' at position 1
age[2]=34;    //value 34 stored in array 'age' at position 2
age[3]=3;     //value 3 stored in array 'age' at position 3
age[4]=4;     //value 4 stored in array 'age' at position 4
```

- Example: Program to illustrate assigning values to one-dimensional array.

```
#include<stdio.h>
void main()
{
        int age[5];
        age[0]=2;
        age[1]=4;
        age[2]=34;
        age[3]=3;
        age[4]=4;

        printf("Value in array age[0] : %d \n", age[0]);
        printf("Value in array age[1] : %d \n", age[1]);
        printf("Value in array age[2] : %d \n", age[2]);
        printf("Value in array age[3] : %d \n", age[3]);
        printf("Value in array age[4] : %d ",age[4]);
}
```

Output:

```
Value in array age[0] :2
Value in array age[1] :4
Value in array age[2] :34
Value in array age[3] :3
Value in array age[4] :4
```


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Reading/Writing to One-Dimensional Array

- For ex:

```
int age[5]
scanf("%d", &age[0]);       //read data from keyboard into array age at position 0.
scanf("%d", &age[1]);       //read data from keyboard into array age at position 1.
scanf("%d", &age[2]);       //read data from keyboard into array age at position 2.
scanf("%d", &age[3]);       //read data from keyboard into array age at position 3.
scanf("%d", &age[4]);       //read data from keyboard into array age at position 4.
```

- In general, to read 5 values, we can write as follows:

```
for(i=0;i<5;i++)
{
        scanf("%d", &age[i]);
}
```

- Similarly, to display 5 elements stored in the array, we can write:

```
for(i=0;i<5;i++)
{
        printf("%d", age[i]);
}
```

- Example: Program to illustrate reading/writing to one-dimensional array.

```
#include<stdio.h>
void main()
{
        int age[5], i;

        printf("enter 5 numbers:\n”);
        for(i=0;i<5;i++)
        {
                scanf("%d", &age[i]);
        }

        for(i=0;i<5;i++)
        {
                printf("Value in array age[%d] : %d \n ", i, age[i]);
        }
}
```

Output:

```
enter 5 numbers:
2 4 34 3 4
Value in array age[0] :2
Value in array age[1] :4
Value in array age[2] :34
Value in array age[3] :3
Value in array age[4] :4
```


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MULTI DIMENSIONAL ARRAYS

- Arrays with two or more dimensions are called multi dimensional arrays.
- For ex:

  int matrix[2][3];
- The above code can be pictorially represented as shown below:

	col 1	col 2	col 3
row 1	a[0][0]	a[0][1]	a[0][2]
row 2	a[1][0]	a[1][1]	a[1][2]

Figure: Multidimensional Arrays

- A two dimensional array is used when elements are arranged in a tabular fashion.
- Here, to identify a particular element, we have to specify 2 indices:

  → First index identifies the row number of the element and

  → Second index identifies the column number of the element

Declaration of Two Dimensional Arrays

- The syntax is shown below:

  data_type array_name[array_size][array_size];
- For ex:

  int matrix[2][3];
- The above code can be pictorially represented as shown below:

	col 1	col 2	col 3
row 1			
row 2			

Initialization of Two Dimensional Arrays

- For ex:

  int matrix[2][3]= {1, 23, 11, 44, 5, 6};
- The above code can be pictorially represented as shown below:

	col 1	col 2	col 3
row 1	1	23	11
row 2	44	5	67

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Reading and Writing Two Dimensional Arrays

- For ex:

```
int matrix[2][3]
scanf("%d", &matrix[0][0]); //read data from keyboard into matrix at row=0 col=0.
scanf("%d", &matrix[0][1]); //read data from keyboard into matrix at row=0 col=1.
scanf("%d", &matrix[0][2]); //read data from keyboard into matrix at row=0 col=2.
scanf("%d", &matrix[1][0]); //read data from keyboard into matrix at row=1 col=0.
scanf("%d", &matrix[1][1]); //read data from keyboard into matrix at row=1 col=1.
scanf("%d", &matrix[1][2]); //read data from keyboard into matrix at row=1 col=2.
```

- In general, to read 6 values, we can write:

```
for(i=0;i<2;i++)
{
        for(j=0;j<3;j++)
        {
                scanf("%d", &matrix[i][j]);
        }
}
```

- Similarly to display 6 elements stored in the matrix, we can write:

```
for(i=0;i<2;i++)
{
        for(j=0;j<3;j++)
        {
                printf("%d ", matrix[i][j]);
        }
}
```

- Example: Program to illustrate reading/writing to two-dimensional array.

```
#include<stdio.h>
void main()
{
        int matrix[2][3], i, j;
        printf("enter elements of 2*3 matrix: \n”);
        for(i=0;i<2;i++)
        {
                for(j=0;j<3;j++)
                {
                        scanf("%d", &matrix[i][j]);
                }
        }
        printf("elements of the matrix are: \n”);
        for(i=0;i<2;i++)
        {
                for(j=0;j<3;j++)
                {
                        printf("%d  \t", matrix[i][j]);
                }
                printf(“\n”);
        }
}
```

Output:

```
enter elements of 2*3 matrix:
1    23  11
44   5    6
elements of the matrix are:
1         23       11
44        5        6
```


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ARRAYS AND FUNCTIONS

• The arrays can be passed to functions using two methods:

1) Passing Individual Elements of Array

• All array-elements can be passed as individual elements to a function like any other variable.

• The array-element is passed as a value parameter i.e. any change in the formal-parameter will not affect the actual-parameter i.e. array-element.

• Example: Program to illustrate passing individual elements of an array to a function.

```
#include<stdio.h>
void display(int a)
{
        printf("%d", a);
}

void main()
{
        int c[3]={2,3,4};
        display(c[2]);      //Passing array-element c[2] only.
}
```

Output:

```
4
```

2) Passing the Whole Array

• Here, the parameter passing is similar to pass by reference.

• In pass by reference, the formal parameters are treated as aliases for actual parameters.

• So, any changes in formal parameter imply there is a change in actual parameter.

• Example: Program to find average of 6 marks using pass by reference.

```
#include <stdio.h>
void average(int m[])
{
    int i ,avg;
    int avg, sum=0;
    for(i=0;i<6;i++)
    {
          sum= sum+ m[i];
    }
    avg =(sum/6);
    printf("aggregate marks= %d ", avg);
}

void main()
{
    int m[6]={60, 50, 70, 80, 40, 80, 70};
    average(m);      // Only name of array is passed as argument
}
```

Output:

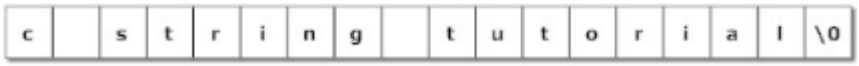
```
aggregate  marks= 70
```


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STRINGS

- Array of character are called strings.
- A string is terminated by null character /0.
- For ex:

  "c string tutorial"
- The above string can be pictorially represented as shown below:



STRING VARIABLE

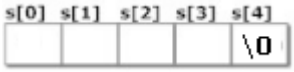
- There is no separate data type for strings in C.
- They are treated as arrays of type 'char'.
- So, a variable which is used to store an array of characters is called a string variable.

Declaration of String

- Strings are declared in C in similar manner as arrays.

  Only difference is that, strings are of 'char' type.
- For ex:

  char s[5]; //string variable name can hold maximum of 5 characters including NULL character
- The above code can be pictorially represented as shown below:

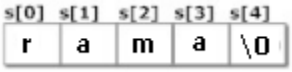


Initialization of String

- For ex:

  char s[5]={'r', 'a', 'm', 'a' };

  char s[9]="rama";
- The above code can be pictorially represented as shown below:



- Example: Program to illustrate the initialization of string.

```
#include<stdio.h>
void main()
{
        char s[5]={'r','a','m','a'};        //or char s[5]="rama";
        printf("Value in array s[0] : %c \n", s[0]);
        printf("Value in array s[1] : %c \n", s[1]);
        printf("Value in array s[2] : %c \n", s[2]);
        printf("Value in array s[3] : %c ", s[3]);
}
```

Output:

```
Value in array s[0] : r
Value in array s[1] : a
Value in array s[2] : m
Value in array s[3] : a
```


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Reading & Printing Strings

• The strings can be read from the keyboard and can be displayed onto the monitor using following 2 formatted functions:

Formatted input function: scanf()
Formatted output function: printf()

• Example: Program to illustrate the use of scanf() and printf().

```
#include<stdio.h>
void main()
{
        char name[10];
        printf("enter your name: \n");
        scanf("%s", name);
        printf("welcome: ");
        printf("%s", name);
}
```

Output:

```
enter your name:
rama
welcome: rama
```

STRING INPUT/OUTPUT FUNCTIONS: gets, puts

• The strings can be read from the keyboard and can be displayed onto the monitor using following 2 unformatted functions:

Unformatted input function: gets()
Unformatted output function: puts()

• Example: Program to illustrate the use of gets() and puts().

```
#include<stdio.h>
void main()
{
        char name[10];
        printf("enter your name: \n");
        gets(name);                    //same as scanf("%s", name);
        printf("welcome: ");
        puts(name);                    //same as printf("%s", name);
}
```

Output:

```
enter your name:
rama
welcome: rama
```


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STRING MANIPULATION FUNCTIONS FROM THE STANDARD LIBRARY

- Strings are often needed to be manipulated by programmer according to the need of a problem.
- All string manipulation can be done manually by the programmer but, this makes programming complex and large.
- To solve this, the C supports a large number of string handling functions.
- There are numerous functions defined in <string.h> header file.

S.N.	Function & Purpose
1	**strcpy(s1, s2);** Copies string s2 into string s1.
2	**strcat(s1, s2);** Concatenates string s2 onto the end of string s1.
3	**strlen(s1);** Returns the length of string s1.
4	**strcmp(s1, s2);** Returns 0 if s1 and s2 are the same; less than 0 if s1<s2; greater than 0 if s1>s2.
5	**strchr(s1, ch);** Returns a pointer to the first occurrence of character ch in string s1.
6	**strstr(s1, s2);** Returns a pointer to the first occurrence of string s2 in string s1.

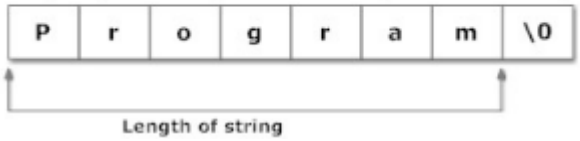
strlen()

- This function calculates the length of string. It takes only one argument, i.e., string-name.
- The syntax is shown below:

```
temp_variable = strlen(string_name);
```

```
char c[]={'P', 'r', 'o', 'g', 'r', 'a', 'm', '\0'};
temp=strlen(c);
```

Then, temp will be equal to 7 because, null character '\0' is not counted.



- Example: Program to illustrate the use of strlen().

```
#include<string.h>
#include<stdio.h>
void main()
{
   char c[20];
   int len;
   printf("Enter string whose length is to be found:");
   gets(c);
   len=strlen(c);
   printf("\n Length of the string is %d ", len);
}
```

Output:

```
Enter string whose length is to be found: program
Length of the string is 7
```


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strcpy()

• This function copies the content of one string to the content of another string. It takes 2 arguments.
• The syntax is shown below:

strcpy(destination,source);

where source and destination are both the name of the string.

• Example: Program to illustrate the use of strcpy().

```
#include<string.h>
#include<stdio.h>
void main()
{
   char src[20],dest[20];
   printf("Enter string: ");
   gets(src);
   strcpy(dest, src);   //Content of string src is copied to string dest
   printf("Copied string: ");
   puts(dest);
}
```

Output:

```
Enter string: vtunotesbysri
Copied string: vtunotesbysri
```

strcat()

• This function joins 2 strings. It takes two arguments, i.e., 2 strings and resultant string is stored in the first string specified in the argument.
• The syntax is shown below:

strcat(first_string,second_string);

• Example: Program to illustrate the use of strcat().

```
#include <stdio.h>
#include <string.h>
void main()
{
  char str1[10], str2[10};
  printf("Enter First String:");
  gets(str1);
  printf("\n Enter Second String:");
  gets(str2);
  strcat(str1,str2);    //concatenates str1 and str2 and
  printf("\n Concatenated String is ");
  puts(str1);           //resultant string is stored in str1
}
```

Output:

```
Enter First String: rama
Enter Second String: krishna
Concatenated String is ramakrishna
```


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strcmp()

• This function compares 2 string and returns value 0, if the 2 strings are equal. It takes 2 arguments, i.e., name of two string to compare.

• The syntax is shown below:

temp_varaible=strcmp(string1,string2);

• Example: Program to illustrate the use of strcmp().

```
#include <string.h>
#include<stdio.h>
void main()
{
  char str1[30],str2[30];
  printf("Enter first string: ");
  gets(str1);
  printf("Enter second string: ");
  gets(str2);
  if(strcmp(str1,str2)==0)
     printf("Both strings are equal");
  else
     printf("Strings are unequal");
}
```

Output:

```
Enter first string: rama
Enter second string: rama
Both strings are equal
```


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WRITING SOME USEFUL STRING FUNCTIONS OF OUR OWN

• Example: Program to find length of a string without using strlen().

```
#include<stdio.h>
void main(void)
{
  char str1[25];
  int len=0;
  printf("Enter string whose length is to be found:");
  gets(str1);

  while(str1[len]!='\0')
     len++;                    //here the length of string is calculated.

  printf("Length of the string is %d", len);
}
```

Output:

```
Enter string whose length is to be found: vtunotesbysri
Length of the string is 13
```

• Example: Program to concatenate two strings without using strcat().

```
void main()
{
  char str1[25],str2[25];
  int i=0,j=0;
  printf(" Enter First String:");
  gets(str1);
  printf("\n Enter Second String:");
  gets(str2);
  while(str1[i]!='\0')
     i++;
  while(str2[j]!='\0')
  {
    str1[i]=str2[j];
    j++;
    i++;
  }
  str1[i]='\0';
  printf("\n Concatenated String is ");
  puts(str1);
}
```

Output:

```
Enter First String: rama
Enter Second String: krishna
Concatenated String is ramakrishna
```


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- Example: Program to copy one string into other without using strcpy().

```
#include<stdio.h>
void main()
{
  char src[100], dest[100];
  int i;
  printf("Enter string:");
  gets(src);
  i = 0;
  while (src[i] != '\0')
 {
    dest[i] = src[i];
    i++;
  }
  dest[i] = '\0';
  printf("Copied String ; %s ", dest);
}
```

Output:

```
Enter string : vtunotesbysri
Copied String : vtunotesbysri
```

- Example: Program to compare 2 strings without using strcmp().

```
#include<stdio.h>
void main()
{
   char str1[100],str2[100];
   int i=0,flag=0;
   printf("Enter first string: ");
   scanf("%s",str1);
   printf("Enter second string: ");
   scanf("%s",str2);
   while(str1[i]!='\0' && str2[i]!='\0')
   {
       if(str1[i]!=str2[i])
      {
          flag=1;
          break;
       }
       i++;
   }
   if (flag==0 && str1[i]=='\0' && str2[i]=='\0')
      flag=1;
   else
      flag=0;
   if(flag == 1)
       printf("Both strings are equal");
   else
       printf("Both strings are not equal");
}
```

Output:

```
Enter first string: rama
Enter second string: rama
Both strings are equal
```


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